BACKBONE V2.1
Medulla Oblongata

1	24V	FIELD 24 V
2	24V	Paralelné napájanie
3	GND	FIELD GND
4	GND	FIELD GND
5	5V	FIELD 5 V
6	5V	Paralelné napájanie
7	NC	Rezerva
8	NC	Rezerva
9	24V_LOGIC	LOGIC 24 V
10	24V_LOGIC	Paralelné napájanie
11	LOGIC_GND	Logická zem
12	LOGIC_GND	Logická zem
13	NC	Rezerva
14	NC	Rezerva
15	24V_BATT	Batériová vetva 24 V
16	5V_BATT	Batériová vetva 5 V
17	BATT_GND	Batériová zem
18	BATT_GND	Batériová zem
19	5V_LOGIC	LOGIC 5 V
20	LOGIC_GND	Logická zem
21	NC	Rezerva
22	NC	Rezerva
23	I2C_SCL	I²C hodinový signál
24	I2C_SDA	I²C dátový signál
25	NC	Rezerva pre budúce IRQ
26	NC	Rezerva
27	SPI_SCK	SPI clock
28	LOGIC_GND	Referenčná zem SPI
29	SPI_MOSI	SPI controller → periféria
30	SPI_MISO	SPI periféria → controller
31	SPI_CS0	Výber SPI zariadenia 0
32	SPI_CS1	Výber SPI zariadenia 1
33	SPI_DC	Display Data/Command
34	LOGIC_GND	Referenčná zem SPI
35	NC	Rezerva
36	RS485_GND	Referenčná zem RS-485
37	RS485_A	Diferenciálna linka A
38	RS485_B	Diferenciálna linka B
39	NC	Rezerva
40	NC	Rezerva

H101
M20-6102045

R120
MBB02070Z0000ZC100

MANUS

IN LINE PULL UP ULNA

R102 ERJPA3F4701V
R103 ERJ-3EKF1001V
C102 C0603C103K5RACTU
R104 ERJPA3F4701V
R105 ERJ-3EKF1001V
C103 C0603C103K5RACTU
R106 ERJPA3F4701V
R107 ERJ-3EKF1001V
C104 C0603C103K5RACTU
R108 ERJPA3F4701V
R109 ERJ-3EKF1001V
C105 C0603C103K5RACTU

LOGIC_3V3, INDEX, POLLEX, DIGITUS_MEDIUS, DIGITUS_ANULARIS, LOGIC_GND

CN101
1990067

DIGITUS MINIMUS

0-30V OUT LINE RADIUS

Pull-down na GPIO
Obmedzenie prúdu LED

R112	ERJ2RKF4702X	OS_Trapenzium	U101 AQY211EH
R110	ERJ3EKF3300V		
R111	ERJ2RKF4702X	OS_Trapezoideum	U102 AQY211EH
R113	ERJ3EKF3300V		
R114	ERJ2RKF4702X	OS_Capitatum	U104 AQY211EH
R115	ERJ3EKF3300V		
R116	ERJ2RKF4702X	OS_Hamatum	U105 AQY211EH
R117	ERJ3EKF3300V		

D101 SMBJ26CA_C83332
D102 SMBJ26CA_C83332
D103 SMBJ26CA_C83332
D104 SMBJ26CA_C83332

CN102
1990067

ESP32-S3-WROOM-1U-N16R8

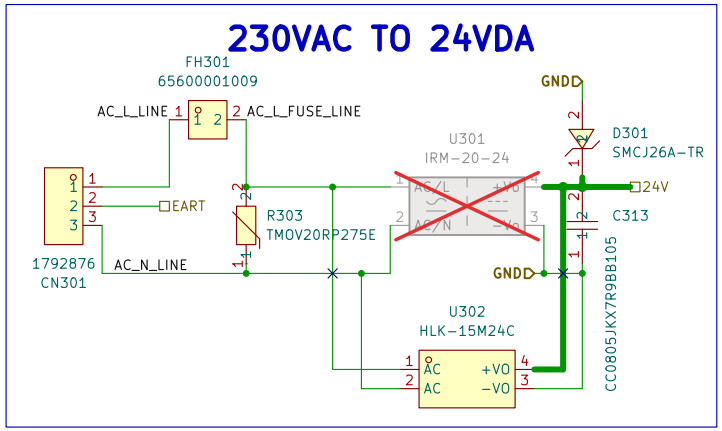
ADC1, ADC2, DI1_LOGIC, DI2_LOGIC, DI3_LOGIC, DI4_LOGIC, DI_AUX1, DO1_CMD, DO2_CMD, DO3_CMD, DO4_CMD, DO_FAULT_INT, ESP_BOOT, RS485_A, RS485_B, LOGIC_GND, LOGIC_3V3, I2C_SCL, I2C_SDA, JTAG TDI, JTAG TMS, RGB_DATA, SPI_CS1, SPI_CS2, SPI_MISO, SPI_MOSI, SPI_SCK, SPI_DC, UART0_RX, UART0_TX, UART_RX, UART_TX, CHASIS, CHASSIS_GND

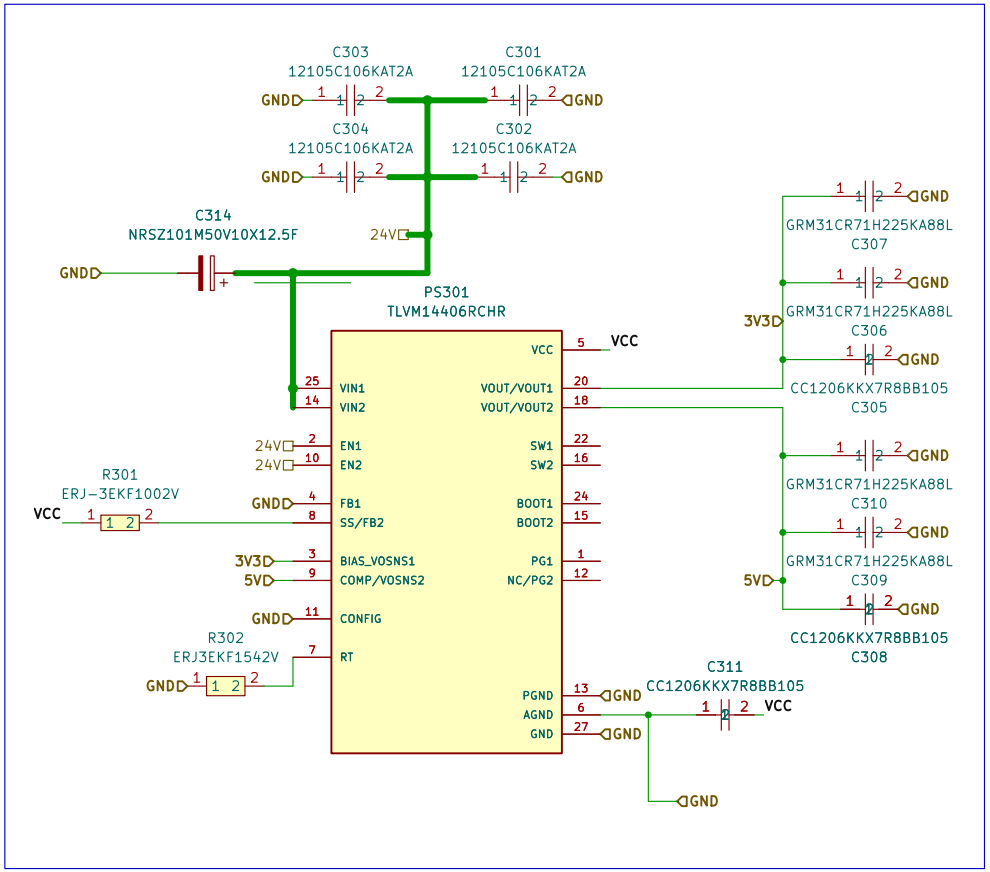
R121 MBB02070Z0000ZC100
R119 MBB02070Z0000ZC100
36V_TO_5V+3V3
C107 DE1E3RA102MA4BN01F
R122 CRCW12061M00FKEA

RGB

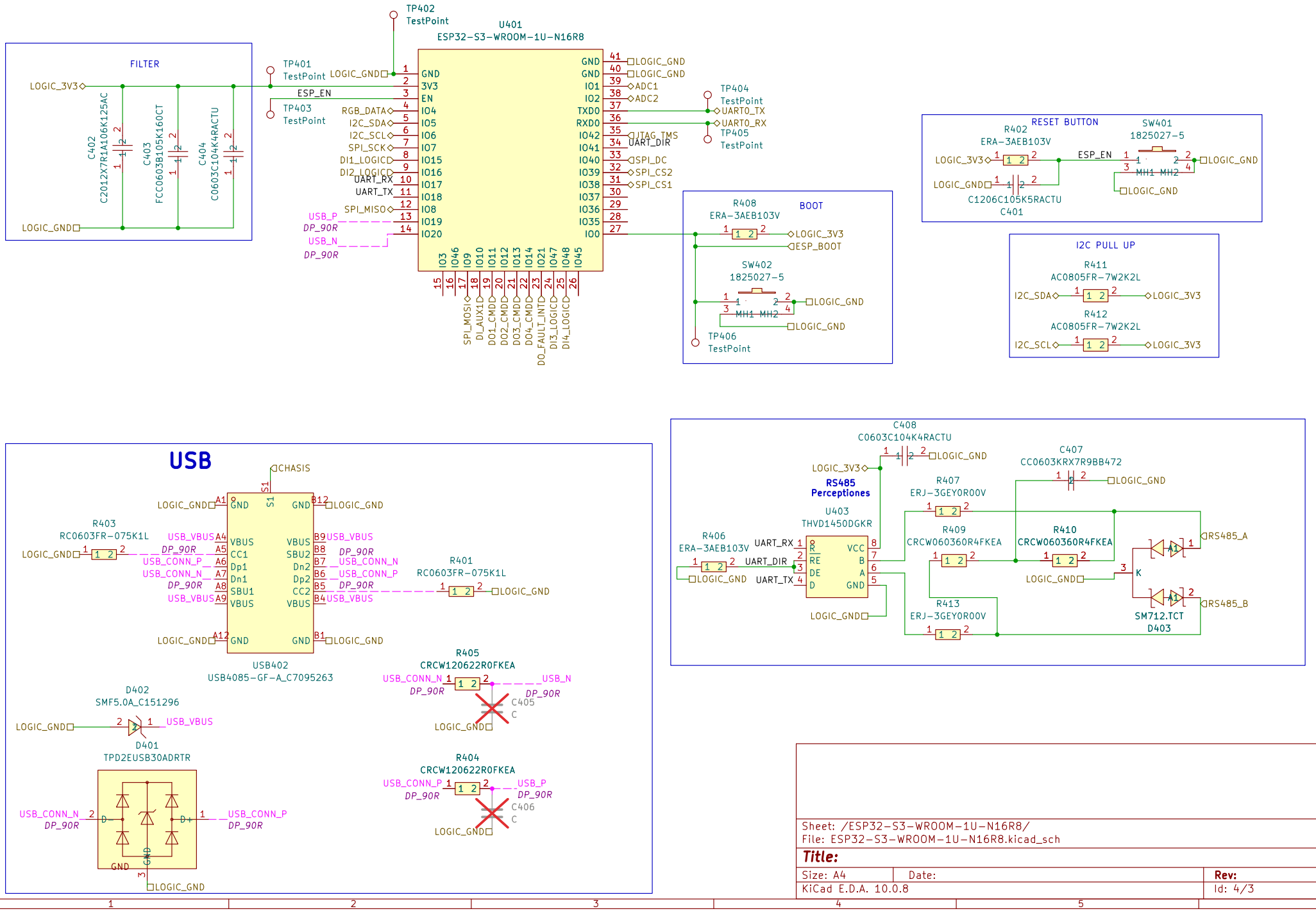
U103 SN74AHCT1G125DBVR
C101 C0805C104K5RACTU_C3018562
R118 ERA-3AEB103V
R101 ERJ3EKF3300V
CN104 1989751

Sheet: /		
File: IOT_LC_PROTOTYPE_V1.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/3





Sheet: /36V_TO_5V+3V3/		
File: 36v to 5v + 3v3.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/3



Sheet: /ESP32-S3-WROOM-1U-N16R8/
File: ESP32-S3-WROOM-1U-N16R8.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.8		Id: 4/3